Cooling Water System — Valve V-104

Operations, Safety & Compliance Manual

Document ID: DOC-CWS-V104-02 | Revision: 3 | System: Cooling Water System | Classification: Internal — Plant Engineering

1. Scope & System Overview

This manual governs the operation, inspection, and maintenance of Gate Valve **V-104**, the primary suction isolation valve for Centrifugal Pump **P-201A** within the Cooling Water System. V-104 is a 6-inch, Class 300 gate valve located immediately upstream of P-201A's suction nozzle. Correct operation of V-104 is a precondition for safe start-up and maintenance of the downstream pump train.

2. Equipment Description

V-104 is a manually-actuated gate valve fitted with a pneumatic quarter-turn override for remote isolation during emergency shutdown sequences. The valve body is carbon steel with a stellite-faced seat, rated for continuous service at up to 150 psi and 90°C process fluid temperature.

3. Inspection & Testing Requirements

- Valve V-104 shall undergo a seat leakage test per API Standard 598 every 90 days; the maximum permissible leakage rate for this hazardous service classification is Rate A (bubble-tight).
- Visually inspect the external body and bonnet of V-104 for corrosion, pitting, or coating degradation at least once every 12 months, and document findings in the corrosion log.
- The pneumatic actuator fitted to V-104 shall be calibrated to achieve a full stroke within 8 seconds at 80 psi supply pressure. Recalibrate the actuator if observed stroke time exceeds 12 seconds during quarterly function testing.

4. Safety Procedures

Before any maintenance activity on V-104 that requires breaking containment, the valve shall be isolated, drained, and vented in accordance with the site's hazardous energy control program. Personnel shall not rely on valve position indication alone to confirm isolation — a physical lockout device must be installed on the handwheel or actuator override before work begins.

5. Compliance References

- The V-104 assembly and its connecting piping fall under the jurisdiction of ASME B31.3, the

Process Piping Code, for design, fabrication, and in-service inspection requirements.

- Isolation and energy-control activities involving V-104 shall comply with OSHA Standard 1910.147, The Control of Hazardous Energy (Lockout/Tagout).